Yuxin Wen

Senior Researcher · Tencent Hunyuan

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Profile

Senior researcher at Tencent Hunyuan working on motion generation and multimodal alignment. Lead author of **HY-Motion 1.0**, the first billion-scale Diffusion Transformer flow-matching model for text-to-motion generation, open-sourced under Tencent Hunyuan. Previously contributed to production-grade lip-sync, facial-expression, and upper-body motion driving systems for flagship games at Tencent. Authored top-venue research in computer vision and machine learning (TPAMI ×3, CVPR ×3, ICML ×1).

Education

Ph.D. in Information and Communication Engineering — *2017–2022*
South China University of Technology (SCUT) — Guangzhou, China

- Advised by Prof. Kui Jia at the Gorilla Lab.

B.Sc. in Information Engineering, Elite Class — *2013–2017*
South China University of Technology (SCUT) — Guangzhou, China

Professional Experience

Senior Researcher — *Oct 2022 – Present*
Tencent Hunyuan / Tencent AI Lab — Shenzhen, China

- **Text-to-Motion Foundation Model (HY-Motion 1.0)**: As lead author of the paper, contributed to the development and open-sourcing of a billion-scale text-to-motion foundation model (GitHub 2k+ stars). Adapted Diffusion Transformer flow matching to motion generation and built a complete training pipeline spanning large-scale pretraining on 3,000+ hours of multi-source data, supervised fine-tuning on curated data, and reinforcement learning from human feedback with reward models, achieving significant gains over existing open-source baselines on instruction following and motion quality.
- **Generative Facial Animation System**: As a core contributor, helped develop a facial animation system that reformulates the traditional driving paradigm as multimodal conditional sequence generation. Built a cross-asset reusable facial generation pipeline; through multi-stage data fitting and model refactoring, tackled transfer to non-standard topology assets.
- **Whole-body Co-driven System with On-device Streaming**: Deeply involved in the design of the system and its on-device streaming architecture. Built a unified multimodal co-generation framework covering face, head, and torso to mitigate semantic discontinuity from isolated single-modality driving; engineered an end-to-end streaming pipeline meeting strict low-latency real-time deployment constraints.

Recognition (Tencent): Outstanding performance rating ×1; Tencent Excellence in R&D Award (company-level) ×2.

Research Intern — *Jun 2021 – Oct 2021*
Alibaba DAMO Academy — Shenzhen, China

- Research internship under the supervision of Prof. Lei Zhang.

Selected Publications

* *first or co-first author.* † *equal contribution.*

1. *__Yuxin Wen__, Qing Shuai, Di Kang, Jing Li, et al. "HY-Motion 1.0: Scaling Flow Matching Models for Text-to-Motion Generation." *Tencent Hunyuan Technical Report*, 2025.
2. Haolin Liu, Xiaohang Zhan, Zizheng Yan, Zhongjin Luo, **Yuxin Wen**, Xiaoguang Han. "Stable-SCore: A Stable Registration-based Framework for 3D Shape Correspondence." *CVPR*, 2025.
3. *Huangzhang Jin[†], **Yuxin Wen**[†], Zihao Wang, Jinjuan Ren, Kui Jia. "Surface Reconstruction from Point Clouds: A Survey and a Benchmark." *TPAMI*, 2022.
4. *Mingyue Yang[†], **Yuxin Wen**[†], Weikai Chen, Yongwei Chen, Kui Jia. "Deep Optimized Priors for 3D Shape Modeling and Reconstruction." *CVPR*, 2021.
5. *__Yuxin Wen__, Jiehong Lin, Ke Chen, C. L. Philip Chen, Kui Jia. "Geometry-Aware Generation of Adversarial Point Clouds." *TPAMI*, 2020.
6. *__Yuxin Wen__[†], Shuai Li[†], Kui Jia. "Towards Understanding the Regularization of Adversarial Robustness on Neural Networks." *ICML*, 2020.
7. Wenbin Zhao[†], Jiabao Lei[†], **Yuxin Wen**, Jianguo Zhang, Kui Jia. "Sign-Agnostic Implicit Learning of Surface Self-Similarities for Shape Modeling and Reconstruction from Raw Point Clouds." *CVPR*, 2021.
8. Shuai Li, Kui Jia, **Yuxin Wen**, Tongliang Liu, Dacheng Tao. "Orthogonal Deep Neural Networks." *TPAMI*, 2019.

Mentorship and Service

- **Mentorship**: Mentored 5–6 research interns at Tencent on motion generation, multimodal alignment, and 3D vision; mentees moved on to top academic and industry positions.
- **Reviewer**: CVPR, ICCV, ECCV, NeurIPS, TPAMI.